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End Semester Examination 2024

Course: B.Tech. (Electronics & Communication Engineering)

Semester: 3rd

Paper Name: Advanced Engineering Mathematics

Paper Code: TMA 310

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks.

Q 1

20 Marks

(a) Define the harmonic function. Check the function given below

$u(x, y) = 2x - x^3 + 3xy^2$ is harmonic or not. If yes, then determine its harmonic conjugate function.

CO1

(b) Find the values of α and β such that the function

$$f(z) = (x^2 + \alpha y^2 - 2xy) + i(\beta x^2 - y^2 + 2xy) \text{ is analytic.}$$

CO1

(c) Determine whether the function $f(z) = \frac{x}{x^2+y^2} + i\frac{y}{x^2+y^2}$ of the complex variable $z = x + iy$ is an analytic function.

CO1

Q 2

20 Marks

(a) Use Cauchy's integral formula to evaluate $\int_C \frac{e^{zt}}{(z^2+1)} dz$, where C is the circle $|z| = 3$.

CO1

(b) Using Newton-Raphson method to compute a real root of the equation

$$x^3 - 5x + 3 = 0 \text{ correct to three decimal places.}$$

CO2

(c) Using Bisection method to compute a real root of the equation

$$2x - \log_{10} x = 7 \text{ correct to four decimal places.}$$

CO2

Q 3**20 Marks**

- (a) Solve $\frac{dy}{dx} = xy$ numerically from $x = 0$ to $x = 1$ using Euler's method with initial condition $y(0) = 1$ and step size (h) of 0.5. Find the absolute value of error in $y(1)$ calculated using analytical and numerical solution (calculate the error using analytical solution as the basis and use two decimal places).

CO3

- (b) Using Newton's forward interpolation formula, Estimate the population for the year 1895 from the table of the population of a town in the decimal census.

Year (x)	1891	1901	1911	1921	1931
Population y (in thousands)	46	66	81	93	101

CO2

- (c) Construct the forward and backward difference table for the data given below:

x	10	15	20	25	30	35
y	19.97	21.51	22.47	23.52	24.65	25.89

CO2**Q 4****20 Marks**

- (a) Evaluate the integral $\int_0^6 \sqrt{x} e^{-x} dx$, using Simpson's one-third rule by taking step-size

$$h = 1.$$

CO3

- (b) An engineering student has a summer job with the forestry service. He measured the tree trunk diameters (x) and related them to the age of the tree (y). The following information was obtained:

$$n = 6, \sum_i x = 21, \sum_i y = 26.4, \sum_i xy = 113.8, \sum_i x^2 = 91, \sum_i y^2 = 142.52$$

Using the least square method and fit the linear estimate is of the form $y = a + bx$.

CO4

- (c) Using Least square method to fit a second-degree curve $y = a + bx + cx^2$ for the

following data:

x	1	4	5	9	14
y	10	5	20	24	21

CO5

Q 5

20 Marks

(a) If the probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} kx^2; & -1 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance, and standard deviation.

CO6

(b) The number of airplane landings at an airport in a minute (X) and their probabilities are given in the following table:

$X = x_i$	0	1	2	3	4	5	6
$p(x_i)$	0.02	0.15	0.22	0.26	0.17	0.14	0.04

Calculate the mean and standard deviation of the random variable X .

CO6

(c) A continuous random variable X has a probability density function

$$f(x) = \frac{3}{4}(2x - x^3), \quad 0 \leq x \leq 1. \text{ Find the value of } a \text{ such that } P(X \leq a) = P(X \geq a).$$

CO6

